

PROJECT APEX

Quantitative Risk Analysis

Version 2.0 | Enhanced Validation & Uncertainty Quantification

Analysis Period: 2021-06-12 to 2026-06-11

1825 observations | 70/30 OOS validation | 500 bootstrap iterations

CONFIDENTIAL

Key Risk Metrics

ANNUALIZED VOL 44.63% 3-4x equity indices	99% DAILY VaR \$80,545 On \$1M portfolio	99% DAILY CVaR \$108,043 Expected tail loss	MAX DRAWDOWN -76.63% 2022-11-21
PERSISTENCE 0.9961 CI: [0.5720, 0.9798]	GPD XI (TAIL) 0.0466 CI: [-0.1355, 0.2325]	OOS VaR 99% PASS Kupiec + Christoffersen + CC	GARCH RESID FAIL Remaining ARCH effects

VERY HIGH RISK | Evidence-based, uncertainty-quantified

v2 enhancements: Formal model selection tests | 70/30 out-of-sample validation | Block bootstrap confidence intervals | Threshold-sensitive EVT | 8 named limitations

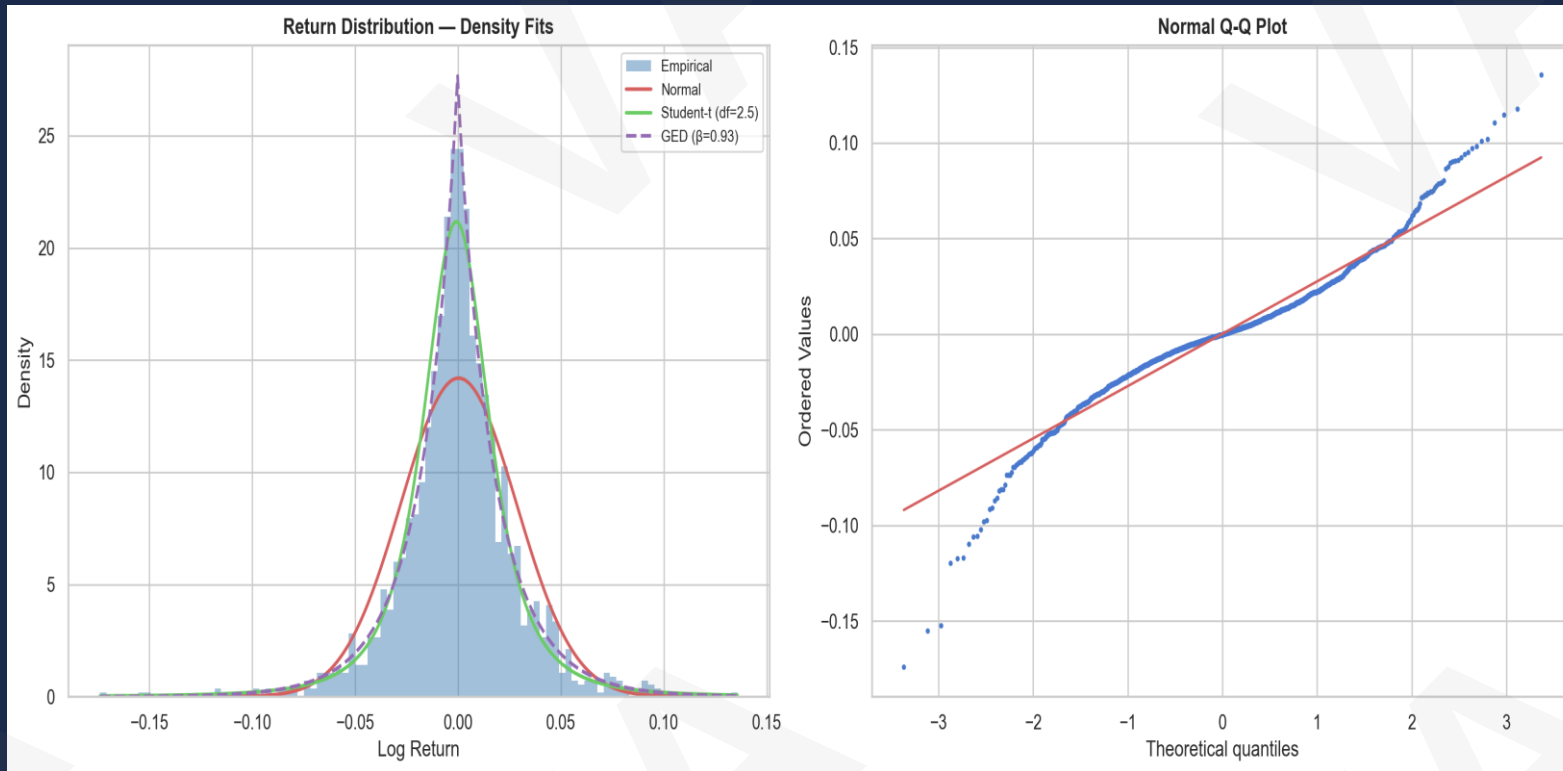
Model Selection: Statistical Evidence

Every model choice backed by formal tests, not visual inspection

Model Selection & Diagnostic Test Summary

Test	Comparison	Statistic	p-value	Result
Likelihood Ratio	Normal vs Student-t	310.75	0.00e+00	Student-t preferred
Vuong	Student-t vs GED	-3.3283	0.0009	GED preferred
Likelihood Ratio	GARCH vs GJR-GARCH	4.5200	0.0335	GJR-GARCH preferred (leverage effect sig
Ljung-Box (sq)	GARCH residuals (lag 10)	26.76	0.0028	FAIL
ARCH-LM	GARCH residuals (lag 10)	27.1003	0.0025	FAIL
Mincer-Zarnowitz	GARCH vol forecast	R2=0.3401	0.0000	FAIL
Diebold-Mariano	GARCH vs GJR-GARCH	0.4134	0.6793	No significant difference between models

Return Distribution: Why Student-t / GED, Not Normal



LR Test: Normal REJECTED ($p < 0.001$)

Student-t df = 2.5

Vuong: GED preferred ($p = 0.0009$)

Kurtosis: 3.97 (Normal = 0)

Practical impact:

Normal VaR 99%: \$65,120

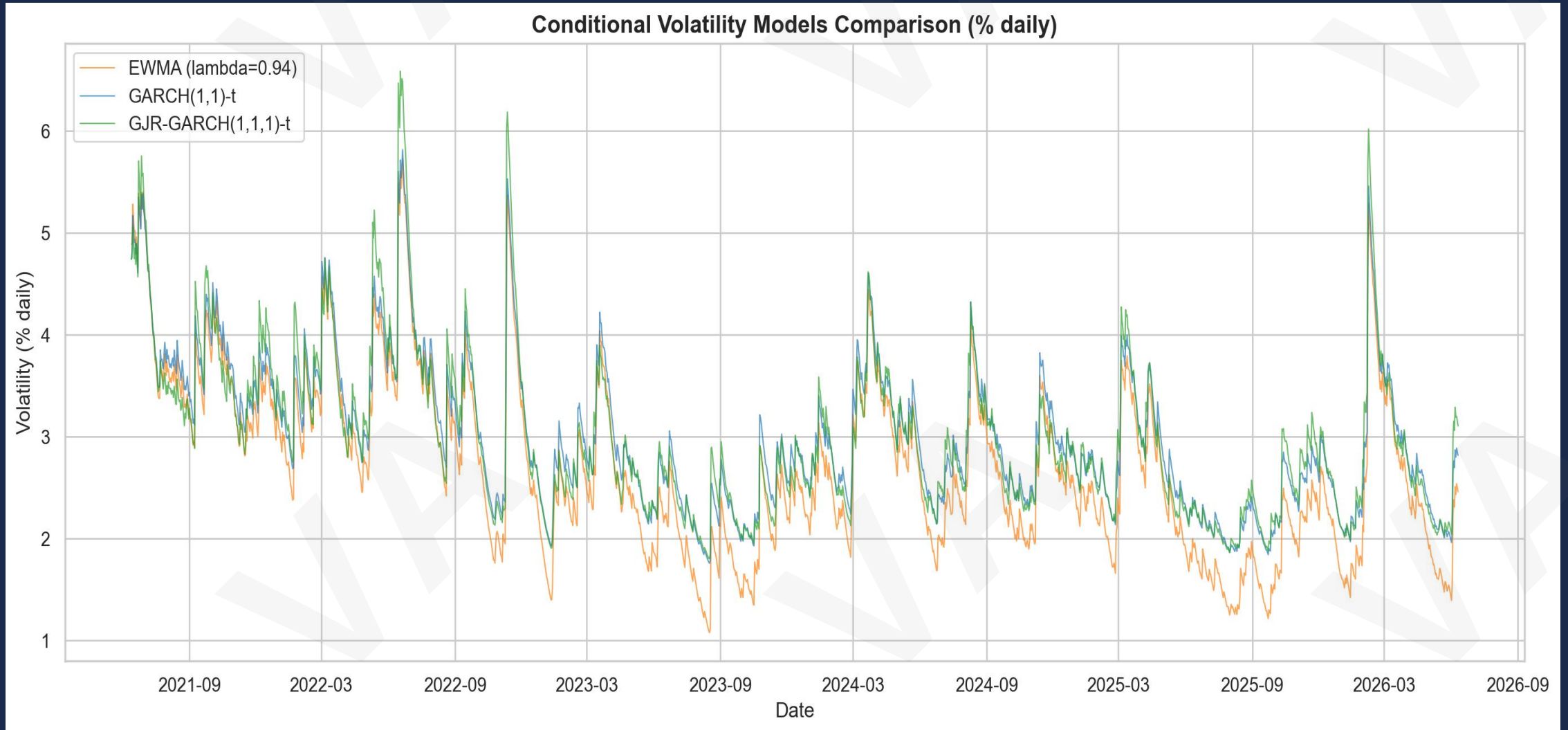
Student-t VaR 99%: \$92,583

Normal underestimates by ~30%

The LR statistic of 310.75 against 1 df makes this one of the clearest model selection decisions in the analysis. Any risk framework using Normal distributions for this asset is materially wrong.

Volatility Models

GJR-GARCH preferred (LR $p=0.0335$) | Residual diagnostics FAIL (ARCH-LM $p=0.0025$) -- honest finding



Out-of-Sample Validation

Training: 1277 obs (to 2024-12-09) | Test: 548 obs (from 2024-12-10)



VaR 99% Backtest (OOS):

Historical: PASS (all 3 tests)

Student-t: PASS (all 3 tests)

Normal 95%: FAIL (Kupiec)

Volatility Forecasts:

MZ R2: 0.3401 (moderate)

MZ Joint Test: FAIL

Forecasts are biased

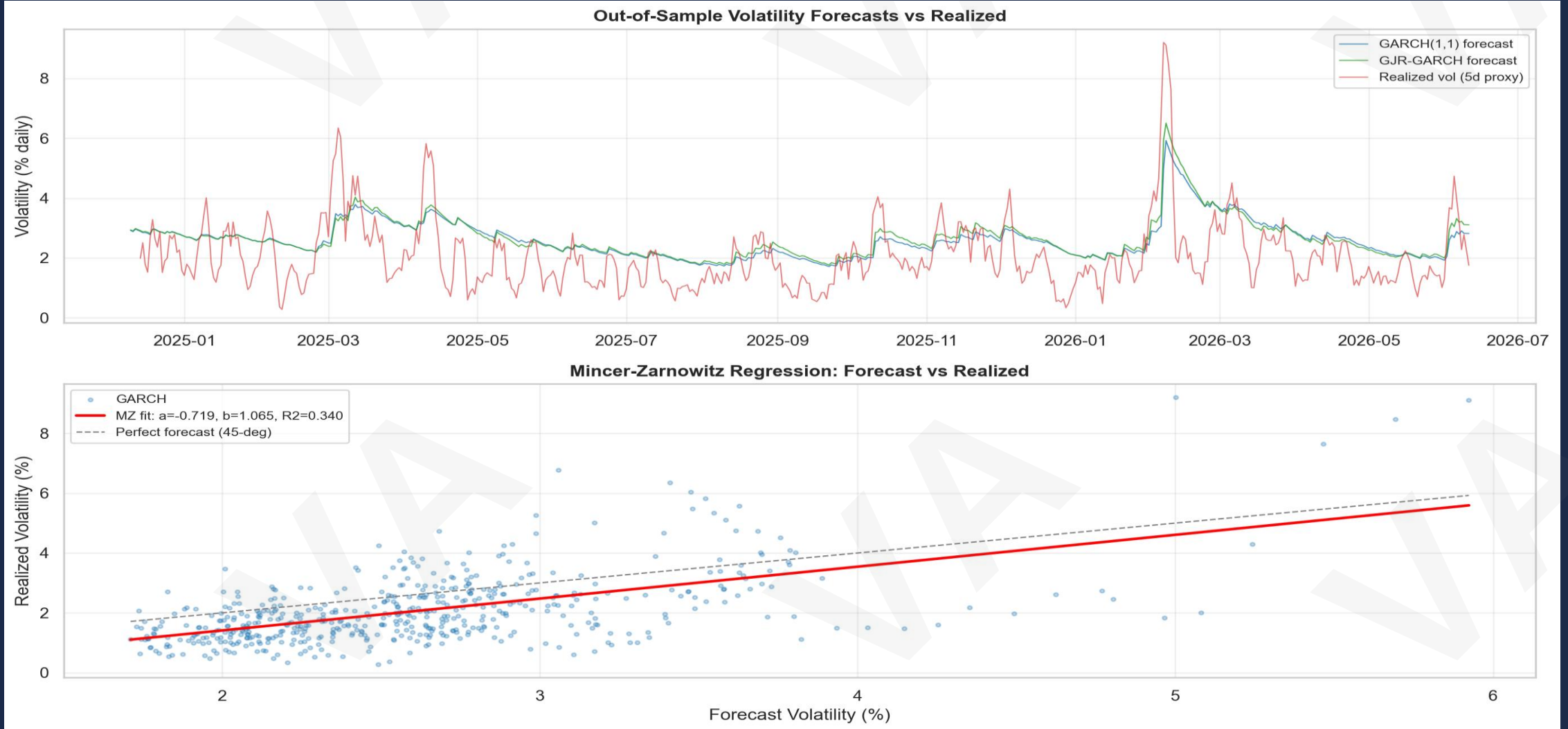
GARCH vs GJR-GARCH:

DM test $p = 0.6793$

No significant OOS difference

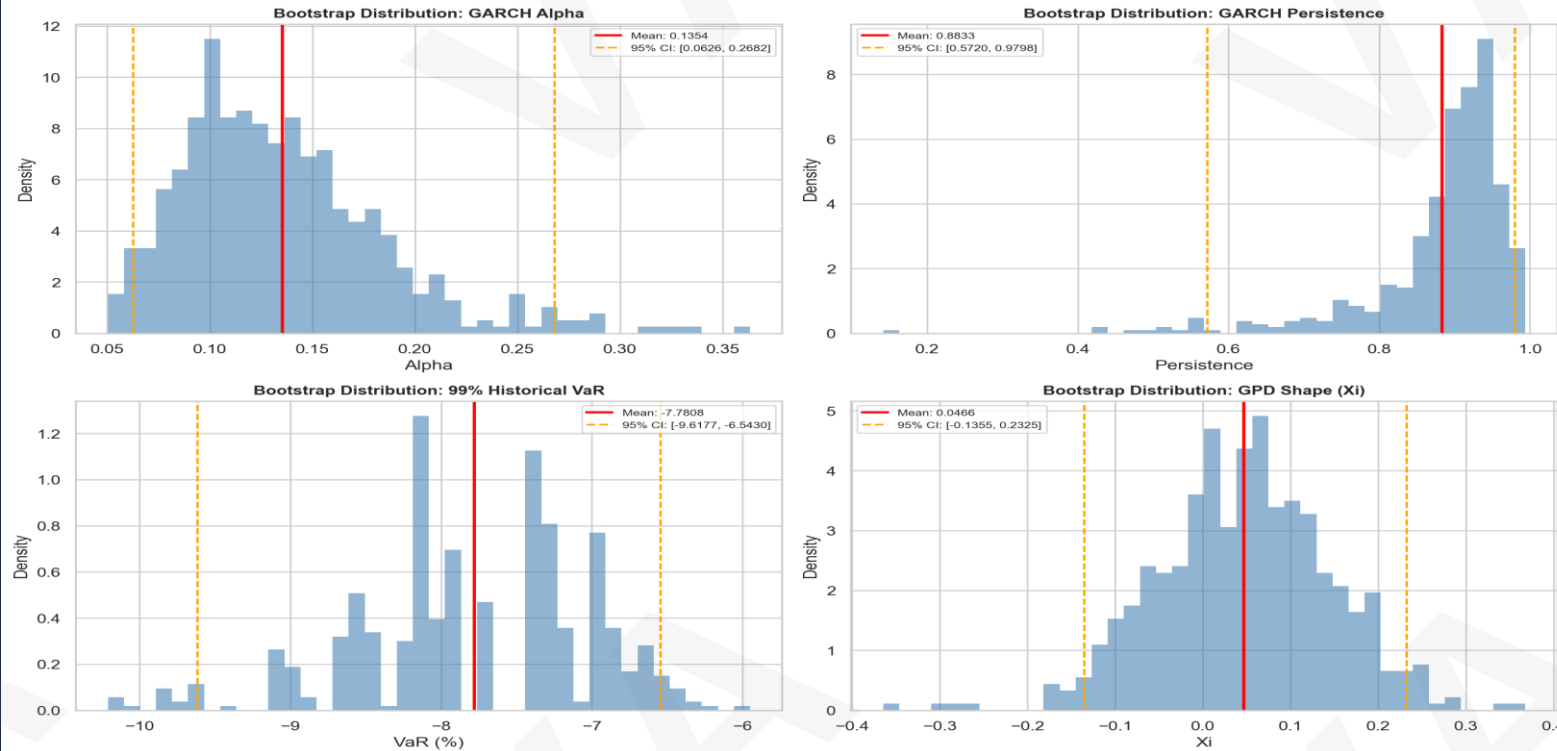
Volatility Forecast Accuracy

MZ: $\alpha=-0.7186$, $\beta=1.0653$, $R^2=0.3401$ | Forecasts explain 34.0% of realized vol but are biased



Parameter Uncertainty: Bootstrap Confidence Intervals

Block Bootstrap Confidence Intervals (500 iterations, block=20 days)



Block Bootstrap (500 iter, block=20d):

GARCH Persistence:

Point: 0.9961

CI: [0.5720, 0.9798]

99% VaR:

CI: [-9.62%, -6.54%]

= \$96,177 to \$65,430

GPD Xi (tail shape):

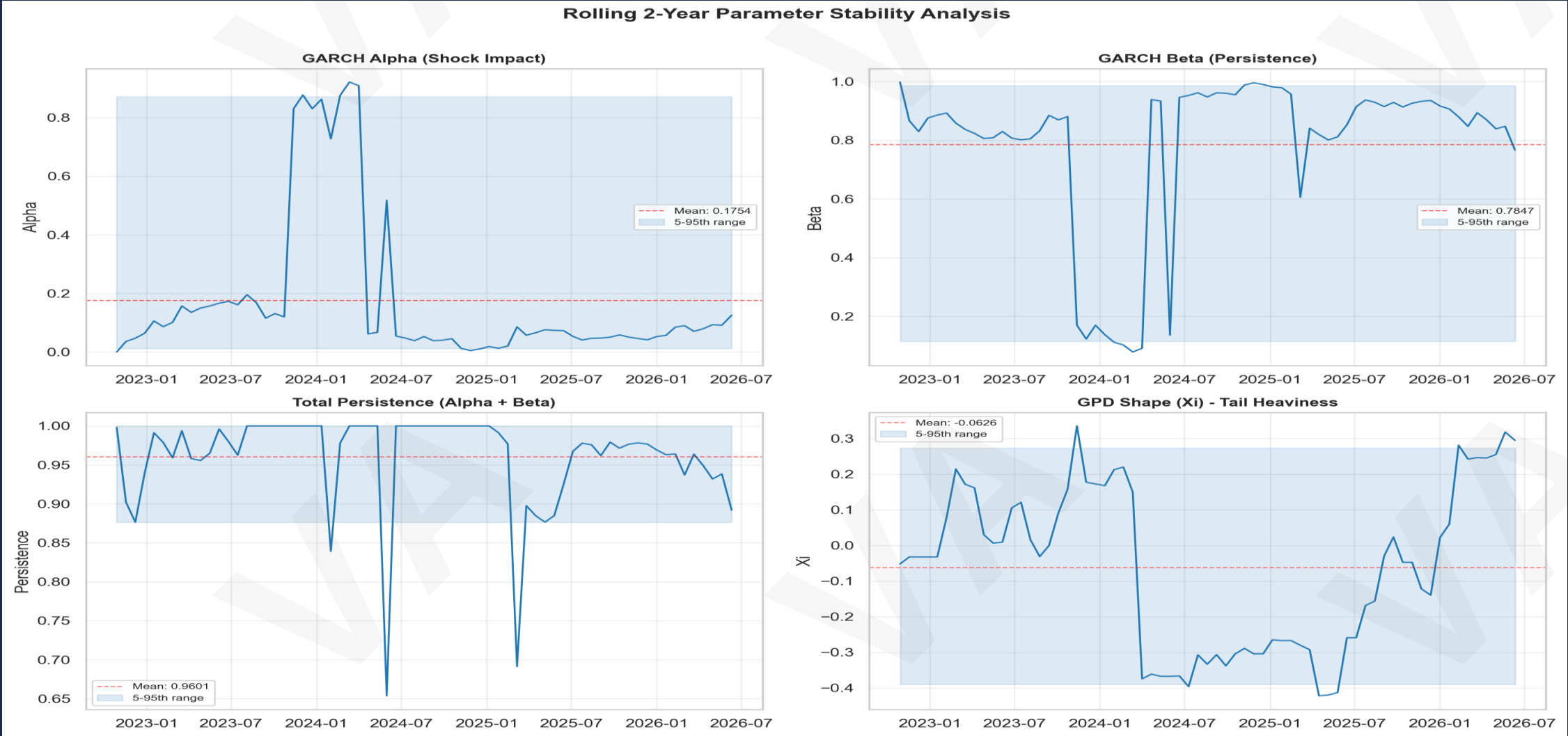
CI: [-0.1355, 0.2325]

Spans neg to pos: UNCERTAIN

n = 92.0 exceedances

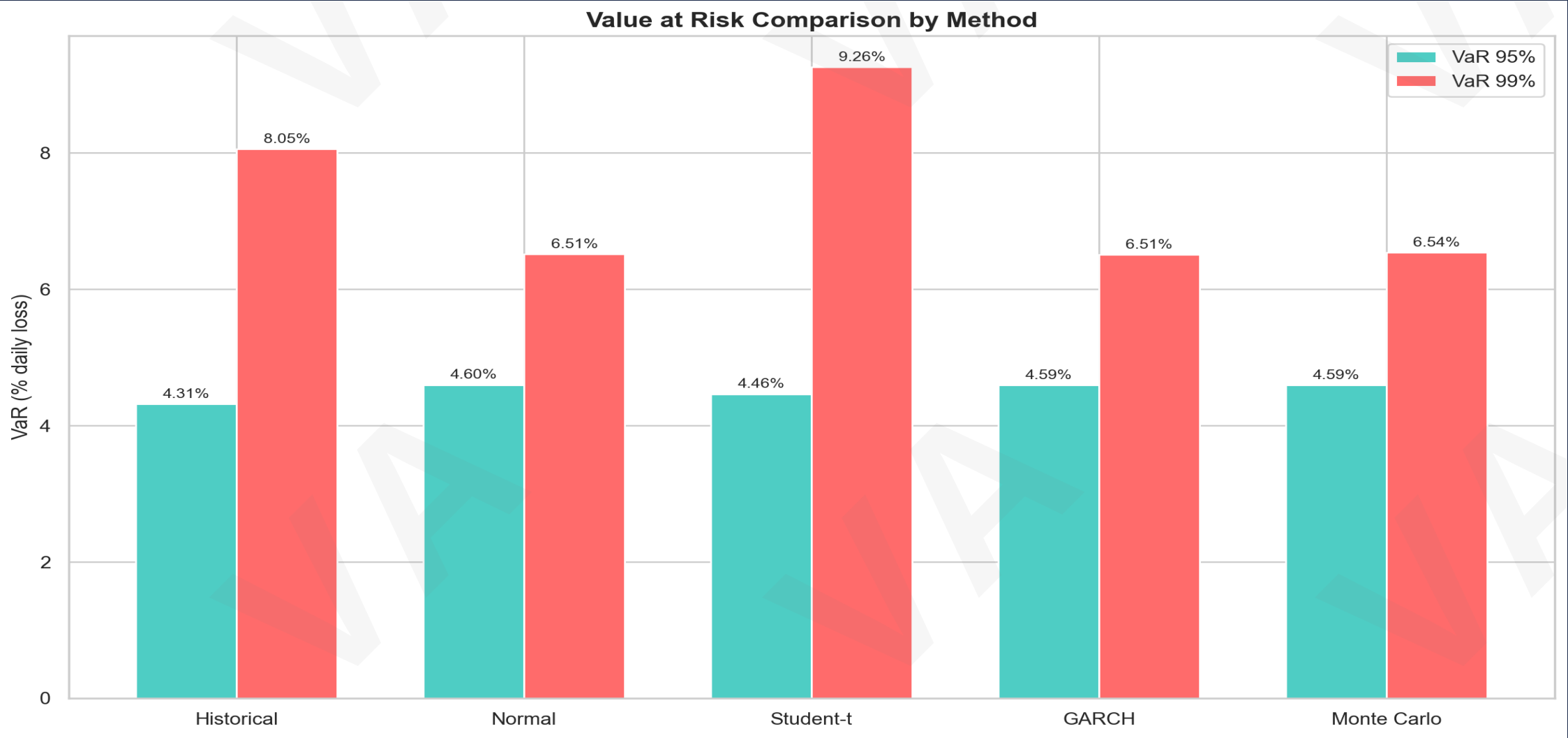
Rolling Parameter Stability (2-Year Windows)

Parameters drift significantly over time -- full-sample estimates are averages across heterogeneous regimes



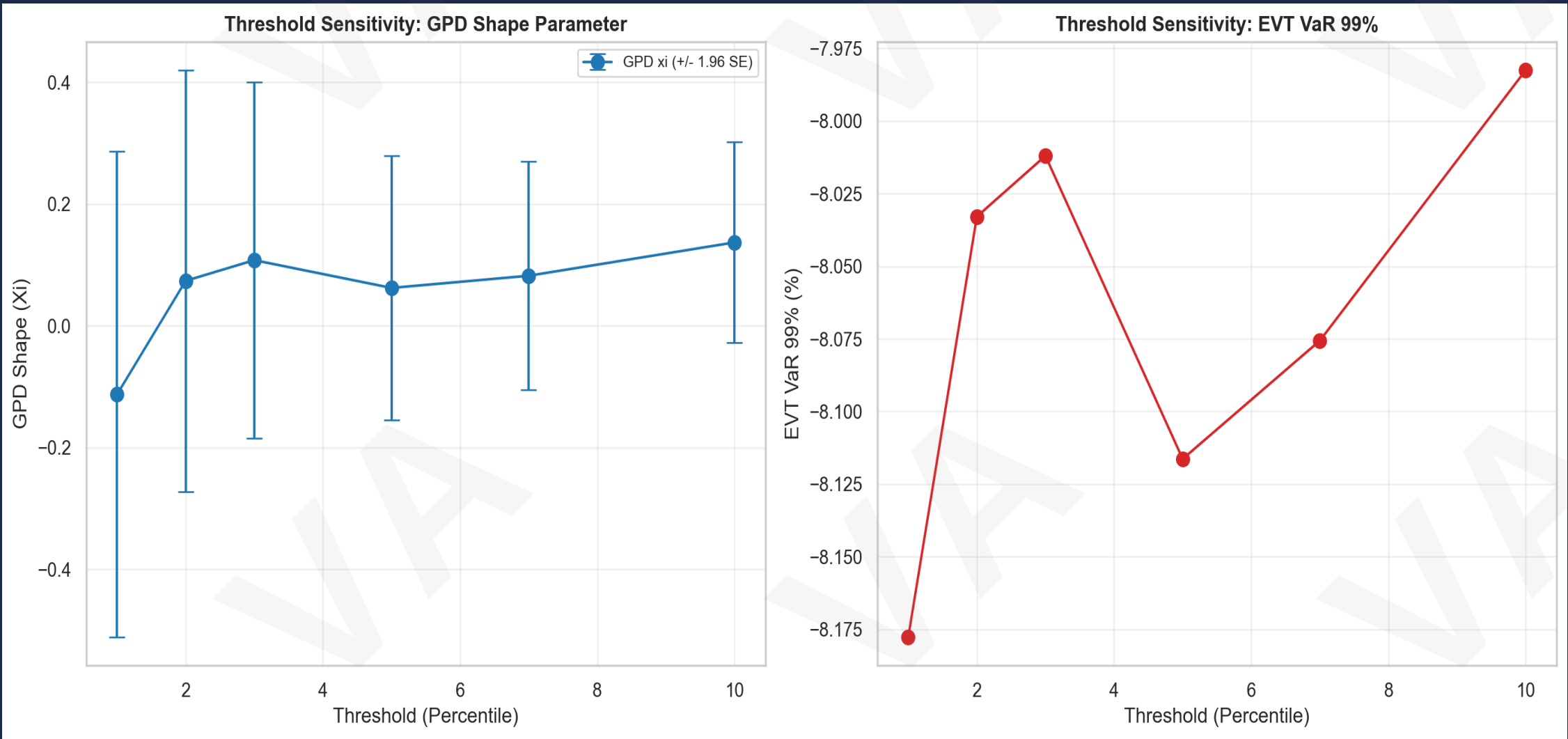
Value at Risk: Method Comparison

Normal VaR underestimates by ~30% at 99% | Bootstrap VaR CI: [-9.62%, -6.54%]



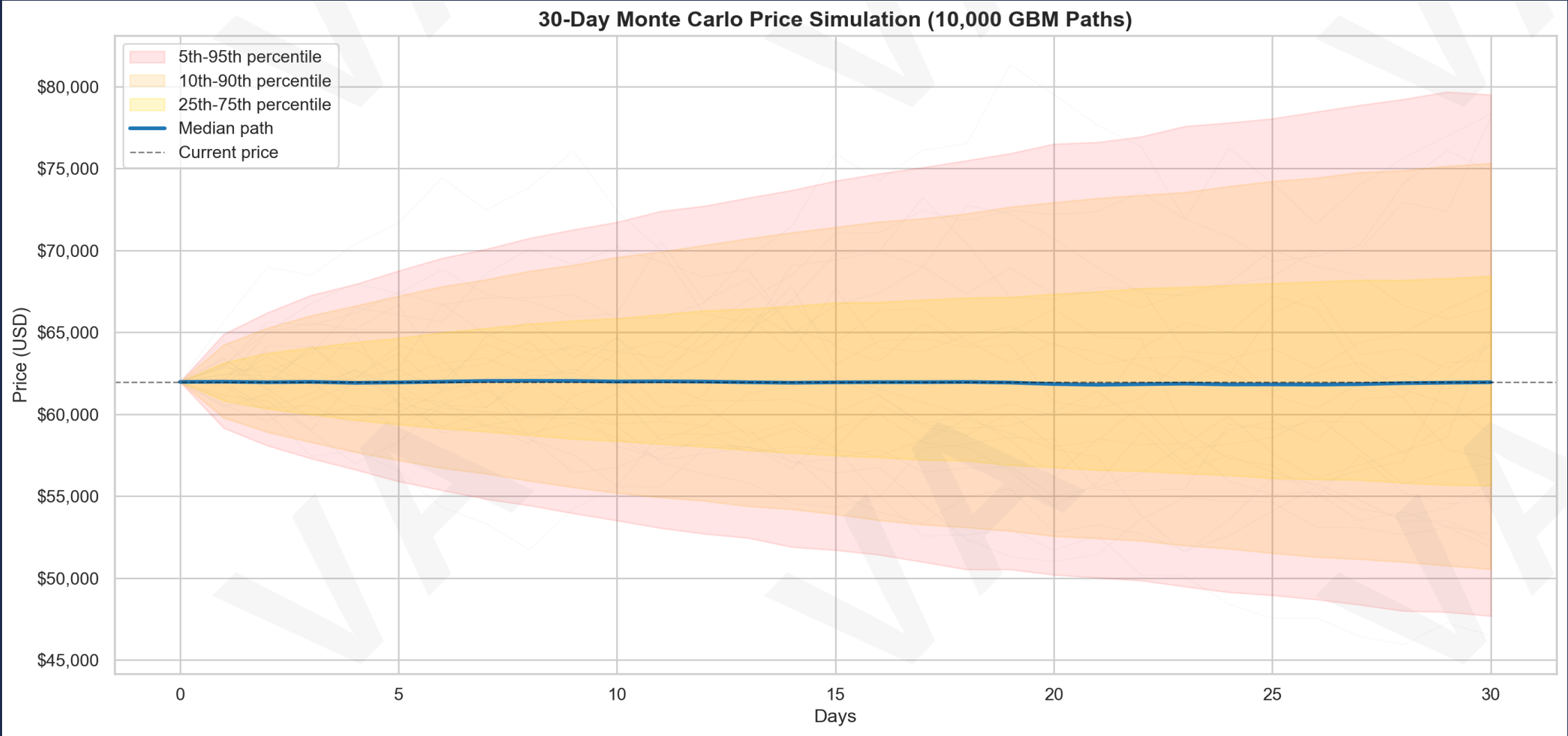
Extreme Value Theory: An Honest Assessment

GPD xi = 0.0466 with 95% CI [-0.1355, 0.2325] | EVT VaR 99% is stable (~8%) despite xi instability | 92.0 tail observations



30-Day Price Simulation

GBM $P(>20\% \text{ loss})$: 8.15% | Jump-diffusion: 15.03% | Jump risk nearly doubles catastrophic loss probability



Known Limitations

A risk framework is only useful if users understand where it breaks

1. STATIONARITY

Markets evolve; parameters drift. Rolling windows show this is not theoretical.

2. REGIME CHANGE

GARCH models smooth transitions, not abrupt shifts. Regime-switching needed.

3. PARAMETER INSTABILITY

Full-sample = average across regimes.
Current conditions may differ materially.

4. TAIL CLUSTERING

EVT assumes IID exceedances. Real tail events cluster, biasing GPD estimates.

5. LIQUIDITY RISK

VaR assumes closing-price liquidation.
Stress spreads multiply actual losses.

6. SINGLE-ASSET SCOPE

No correlation or contagion modeling.
Portfolio risk requires copula/DCC-GARCH.

7. SURVIVORSHIP BIAS

This asset survived. Failed assets are not in the sample.

8. DATA FREQUENCY

Daily data misses intraday risk in a 24/7 market.

Evidence-Based Recommendations

WHAT THE EVIDENCE SUPPORTS:

Use fat-tailed VaR (Student-t/GED)

Normal underestimates by ~30% at 99%

Report CIs, not point estimates

VaR CI: [-9.62%, -6.54%]

OOS VaR backtests PASS at 99%

Models are well-calibrated for tails

No leverage. Max DD = -76.63%

WHAT THE EVIDENCE DOES NOT SUPPORT:

Precise tail quantification

GPD xi CI spans negative to positive

Unbiased vol forecasting

MZ test rejects unbiasedness

Parameter stability

Rolling windows show material drift

Re-estimate quarterly at minimum

PROJECT APEX

Quantitative Risk Analysis v2.0

Next: Multi-Asset Portfolio | Regime-Switching | Intraday Risk